

Exercise 3 - Precipitation Analysis in the Madi Catchment, Nepal

Task 1 - Estimation of Missing Precipitation Data

The objective of the first part of the exercise was to reconstruct missing daily precipitation observations for stations 802 (Khudi Bazar) and 821 (Ghandruk). Six precipitation stations located within or around the Madi catchment were available, with elevations ranging from 856 to 3420 m a.s.l. and annual precipitation ranging from 448 to 3752 mm.

The selection of predictor stations was based primarily on similarities in annual precipitation and elevation, as suggested in the exercise. Stations 811 (Melepatan Pokhara) and 824 (Sikles) were therefore considered the most suitable predictors. Stations 820 (Manang Bot) and 816 (Chame) were not included in the regression because their annual precipitation is substantially lower than that observed at stations 802 and 821, indicating a markedly different precipitation regime.

A multiple linear regression was used to estimate the missing observations. For station 802, the resulting relationship was:

$$P_{82} = 5.093 + 0.0758 P_{811} + 0.3553 P_{824}$$

where precipitation is expressed in mm.

The missing values for station 802 were consequently estimated as:

Date	Estimated P802 (mm)
20/6	16.05
21/6	15.98

For station 821, the corresponding regression equation was:

$$P_{821} = -14.350 + 0.4453 P_{811} + 0.9081 P_{824}$$

resulting in the following estimated values:

Date	Estimated P821 (mm)
20/6	17.26

21/6	19.67
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The regression provided a substantially stronger relationship for station 821 ($R^2 \approx 0.73$) than for station 802 ($R^2 \approx 0.13$). The latter indicates considerable uncertainty in the reconstructed values and highlights the limitations of estimating individual precipitation events from a short observational period. The original exercise table also contains a measured value of 0 mm for station 821 on 22/6; this value was therefore retained rather than replaced.

Task 2 - Wind Correction of Precipitation Measurements

Precipitation measurements from conventional gauges may be underestimated under windy conditions because airflow around the gauge can cause precipitation particles to be deflected away from the collector. In this exercise, all precipitation was assumed to occur as rain, the gauge was assumed to be sheltered, and the same wind speed was applied to all stations.

The correction factor was obtained from the mean rain curve provided with the exercise. The graph expresses the relationship between observed and true precipitation as:

$$F = \frac{P_{\text{observed}}}{P_{\text{true}}}$$

Therefore, the corrected precipitation was calculated as:

$$P_{\text{true}} = \frac{P_{\text{observed}}}{F}$$

The correction factor decreases with increasing wind speed, meaning that the correction becomes progressively larger under stronger wind conditions. Approximate factors were read from the mean rain curve for each observed wind speed.

For example, for 26/6 the recorded wind speed was 1.2 m/s. The corresponding correction factor was approximately 0.96. Thus, for station 811:

$$P_{\text{true}} = \frac{70.2}{0.96} \approx 73.1 \text{ mm}$$

The resulting corrected precipitation dataset is presented below.

Date	Wind (m/s)	802	811	824	821	820	816
19/6	0.0	37.00	16.80	35.60	18.20	4.50	13.50
20/6	0.5	16.38	14.59	28.37	17.61	14.59	0.51
21/6	0.3	16.14	24.85	25.66	19.87	5.25	31.21
22/6	1.5	17.02	0.11	23.40	0.00	5.96	1.60
23/6	2.0	6.52	7.39	3.26	0.00	5.00	2.17
24/6	0.1	1.51	44.32	9.75	3.22	1.71	0.10
25/6	0.5	2.55	22.45	39.18	42.04	0.00	0.00
26/6	1.2	22.92	73.13	10.83	31.98	0.00	4.48

The correction is relatively small at low wind speeds but becomes increasingly relevant as wind speed increases. The corrected values were subsequently used for the areal precipitation calculation.

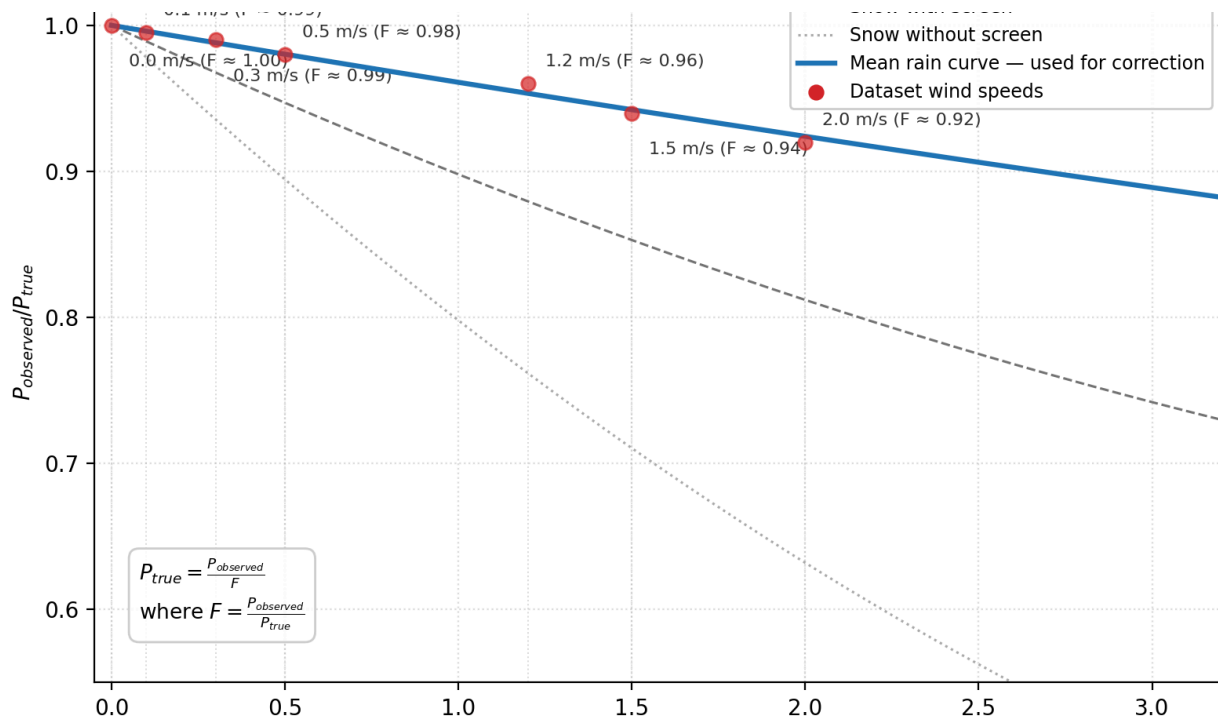


Figure 1. Wind correction relationship used to adjust precipitation measurements for the effects of wind. The correction factors were obtained from the mean rain curve supplied with the exercise.

Task 3 - Areal Precipitation Using the Thiessen Method

The spatial variability of precipitation across the Madi catchment was estimated using the Thiessen polygon method. The method assigns each part of the catchment to the precipitation station that is geographically closest to it. The resulting polygon areas were determined from the supplied Madi catchment map by counting the 1 km² grid squares, as specified in the exercise.

The resulting approximate Thiessen areas and corresponding weighting factors were:

Station	Thiessen area (km ²)	Weight
802	294	0.323
811	123	0.136
824	457	0.503
821	~0	~0
820	~0	~0
816	35	0.038
Total	~909	1.000

The spatial distribution indicates that station 824 contributes the largest proportion of the catchment, accounting for approximately 50.3% of the total area, followed by stations 802 and 811. Stations 821 and 820 do not contribute appreciably to the calculated catchment-average precipitation under the Thiessen construction.

[Figure 13. Thiessen polygon map of the Madi catchment. The original map should be retained as the base layer, with the six precipitation stations clearly labelled and the Thiessen boundaries shown using thin lines. The catchment boundary and the relative area of each polygon should be visually clear. A minimal legend should identify the station numbers. This is probably the most important figure of Exercise 3.]

The areal precipitation for each day was calculated using the area-weighted mean:

$$P_A = \frac{\sum_{i=1}^n P_i A_i}{A}$$

or, equivalently, using the Thiessen weights:

$$P_A = \sum_{i=1}^n w_i P_i$$

where P_i is the corrected precipitation at station i , A_i is the corresponding Thiessen polygon area, and $w_i = \frac{A_i}{A}$ is the area weighting factor.

Using the calculated Thiessen weights, the resulting daily areal precipitation values were:

Date	Areal precipitation (mm)
19/6	32.66
20/6	21.55
21/6	22.68
22/6	17.35
23/6	4.83
24/6	11.39
25/6	23.56
26/6	22.93

The highest catchment-average precipitation occurred on 19/6, with approximately 32.7 mm, whereas the lowest value occurred on 23/6, with approximately 4.8 mm. The areal values integrate the spatial variability observed between the individual gauges and therefore provide a more representative estimate of precipitation over the entire catchment than any single point measurement.

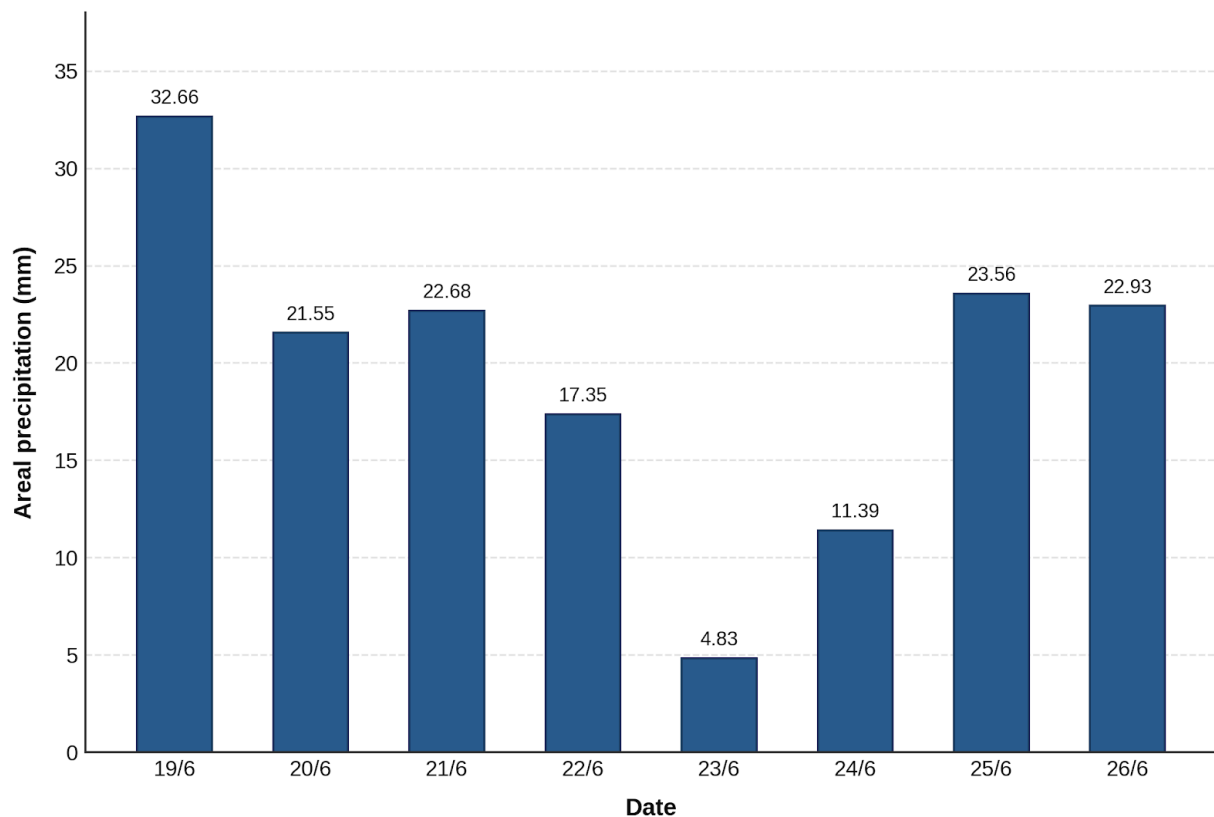


Figure 3. Daily areal precipitation over the Madi catchment calculated using wind-corrected station precipitation and Thiessen polygon weighting for 19–26 June.

Task 4 - Relationship Between Precipitation and Elevation

In Norway, precipitation is commonly assumed to increase with elevation, and elevation-dependent precipitation gradients are frequently used to estimate spatial precipitation variability from point observations. The final part of the exercise evaluates whether this assumption is appropriate for the Madi catchment in Nepal.

The station data do not show a consistent positive relationship between elevation and annual precipitation. Annual precipitation initially increases from the low-elevation stations towards intermediate elevations, reaching a maximum of 3752 mm at Sikles (1820 m). However, precipitation decreases strongly at higher elevations, reaching only 1050 mm at Chame (2680 m) and 448 mm at Manang Bot (3420 m).

The relationship can be illustrated by comparing the two highest-elevation stations with the intermediate-elevation stations:

- Sikles: 1820 m → 3752 mm/year
- Ghandruk: 1960 m → 3207 mm/year
- Chame: 2680 m → 1050 mm/year
- Manang Bot: 3420 m → 448 mm/year

A simple correlation between elevation and annual precipitation for the six stations gives approximately:

$$r = -0.88$$

Although this value should be interpreted cautiously because only six stations are available, it clearly does not support a simple positive precipitation–elevation gradient.

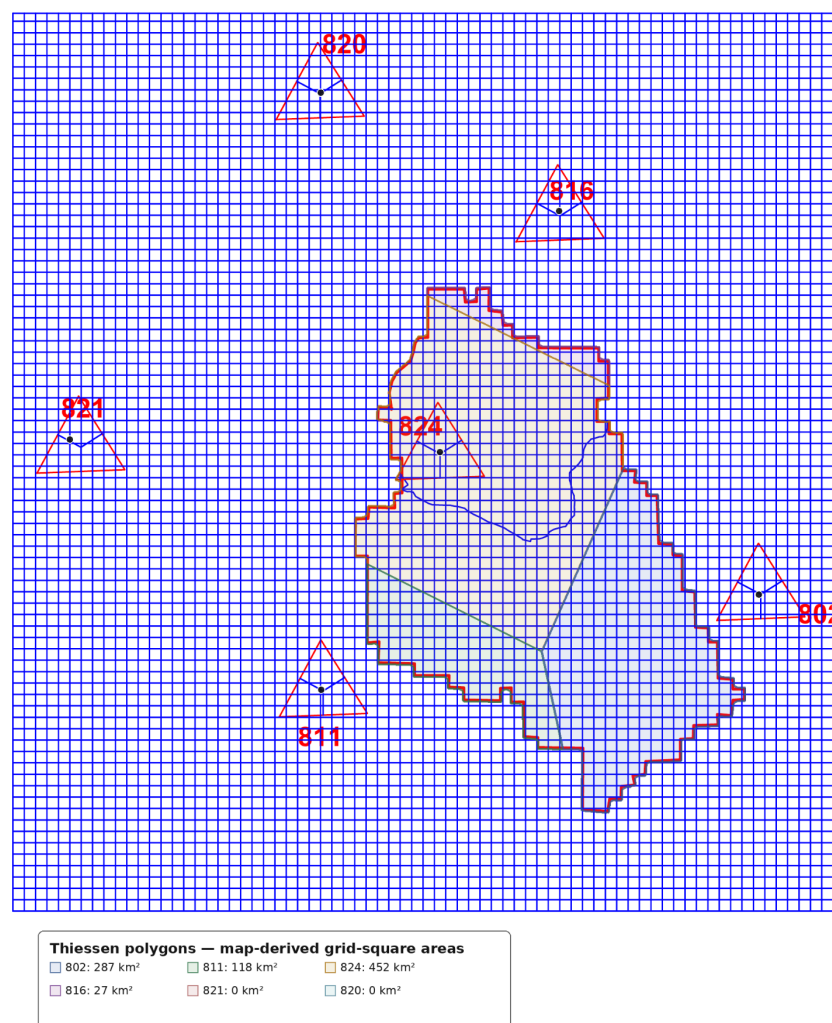


Figure 4. Relationship between station elevation and annual precipitation for the six precipitation stations

surrounding the Madi catchment. The observations do not show a consistent positive precipitation–elevation relationship.

The observed pattern indicates that precipitation in the Madi region is controlled by more than elevation alone. The strong decrease in precipitation at the highest elevations suggests substantial spatial variability in the regional precipitation regime. Consequently, the positive precipitation–elevation relationship commonly assumed in Norway cannot be directly applied to this Nepalese catchment.

For this reason, precipitation estimates in the Madi catchment should rely on the observed spatial distribution of the available stations rather than assuming that precipitation systematically increases with elevation.

